Bureau
International des
Poids et
Mesures

Le système international d'unités
The International System of Units

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Annexe 2
Appendix 2

Valeurs recommandées des fréquences étalons destinées à la mise en pratique de la définition du mètre et aux représentations secondaires de la seconde

Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second

Partie 2.5
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Mercury 199 Ion ($f \approx 1\,065\,\mathrm{THz}$)

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juillet 2013

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Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second
Part 2.5: Mercury 199 Ion ($f \approx 1\,065\,\mathrm{THz}$)

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$^{199}Hg^{+}$ ion, $5d^{10}6s\ ^{2}S_{1/2} - 5d^{9}6s^{2}\ ^{2}D_{5/2}$ unperturbed optical transition

1. Recommended value [1] of the frequency:

$f\left({}^{199}\mathrm{Hg}^{+}\right) = 1\,064\,721\,609\,899\,145.30\ \mathrm{Hz}$

equivalent to

$\lambda\left({}^{199}\mathrm{Hg}^{+}\right) = 281\,568\,867.591\,968\,5\ \mathrm{fm}$

with a relative uncertainty of 1.9×10^{-15}.

This radiation was already endorsed as a secondary representation of the definition of the second.

2. Source data

Adopted value $f(^{199}\mathrm{Hg}^{+}) = 1\,064\,721\,609\,899\,145.30\ \mathrm{Hz}$

$u_c/y = 1.9 \times 10^{-15}$

calculated from

$f(^{199}\mathrm{Hg}^{+})$/Hz	u/Hz	source data
1 064 721 609 899 145.30	0.69	[2]

As this value was issued from only one laboratory the CCTF considered it prudent to estimate a relative standard uncertainty of 1.9×10^{-15}.

References

[1] CIPM Recommendation 1(CI-2013): Updates to the list of standard frequencies.

[2] J.E. Stalnaker, S.A. Diddams, T.M. Fortier, K. Kim, L. Hollberg, J.C. Bergquist, W.M. Itano, M.J. Delany, L. Lorini, W.H. Oskay, T.P. Heavner, S.R. Jefferts, F. Levi, T.E. Parker and J. Shirley: Optical-to-microwave frequency comparison with fractional uncertainty of 10^{-15}, *Appl. Phys. B*, **89**, 167 (2007).